

Practical 3: Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Problem Definition: Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

Key Questions:

1. How does the layout adapt for mobile, tablet, and desktop screens?

Answer:

- The website uses **Bootstrap's responsive grid system** to adapt to different screen sizes.
- **Mobile:** Content is stacked vertically for easy reading and navigation.
- **Tablet:** Elements are arranged with balanced spacing to make better use of the screen.
- **Desktop:** Multiple columns are displayed side by side to utilize the larger screen efficiently.
- Bootstrap breakpoints automatically adjust the layout, ensuring a consistent user experience across all devices.

2. Which layout technique is used and why?

Answer:

Layout Technique Used: Bootstrap Grid System (Flexbox)

- Creates a responsive layout that works on all screen sizes.
 - Divides the page into a **12-column grid**, making content organization easier.
 - Flexbox provides flexible alignment and spacing of elements.
 - Reduces the need for writing complex CSS.
 - Makes the website easier to maintain and update.
3. Are typography, spacing, colors, and contrast readable?

Answer:

Yes. The project follows good UI and readability practices:

- **Typography:** Uses clear and readable font sizes with proper font hierarchy.
- **Spacing:** Appropriate margins and padding are used to avoid a cluttered interface.
- **Colors:** A consistent color scheme is maintained throughout the website.
- **Contrast:** Good contrast between text and background improves readability.
- **Accessibility:** The design is comfortable to read on different devices and screen sizes.

These elements together improve the overall user experience and make the interface visually appealing.

4. Is the framework used consistently without unnecessary inline CSS?

Answer:

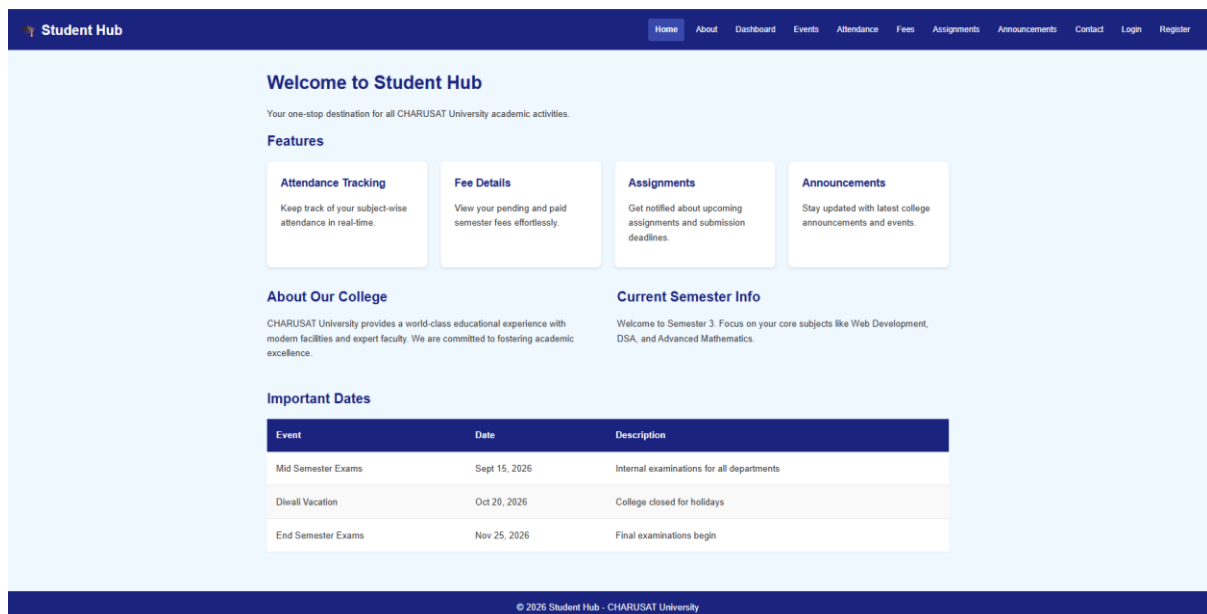
Yes. The project uses Bootstrap consistently throughout the website.

- Bootstrap classes are used for layout, buttons, forms, navigation, and responsive design.
- Most styling is achieved using Bootstrap's built-in utility classes.
- Custom styles, if required, are placed in an **external CSS file** instead of inline CSS.
- Unnecessary inline CSS is avoided, making the code cleaner and easier to maintain.
- This approach improves code reusability, readability, and simplifies future modifications.

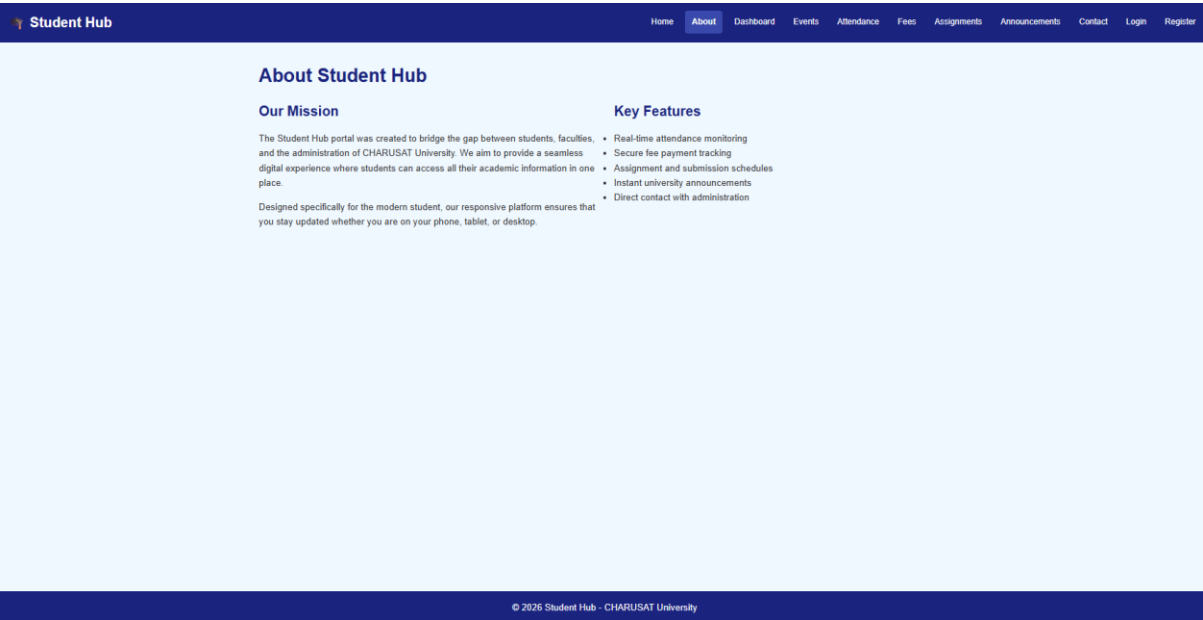
For Codes: <https://github.com/kunjborad21/WebDev-CSE1-A/tree/main/Practical%203>

Webpages:

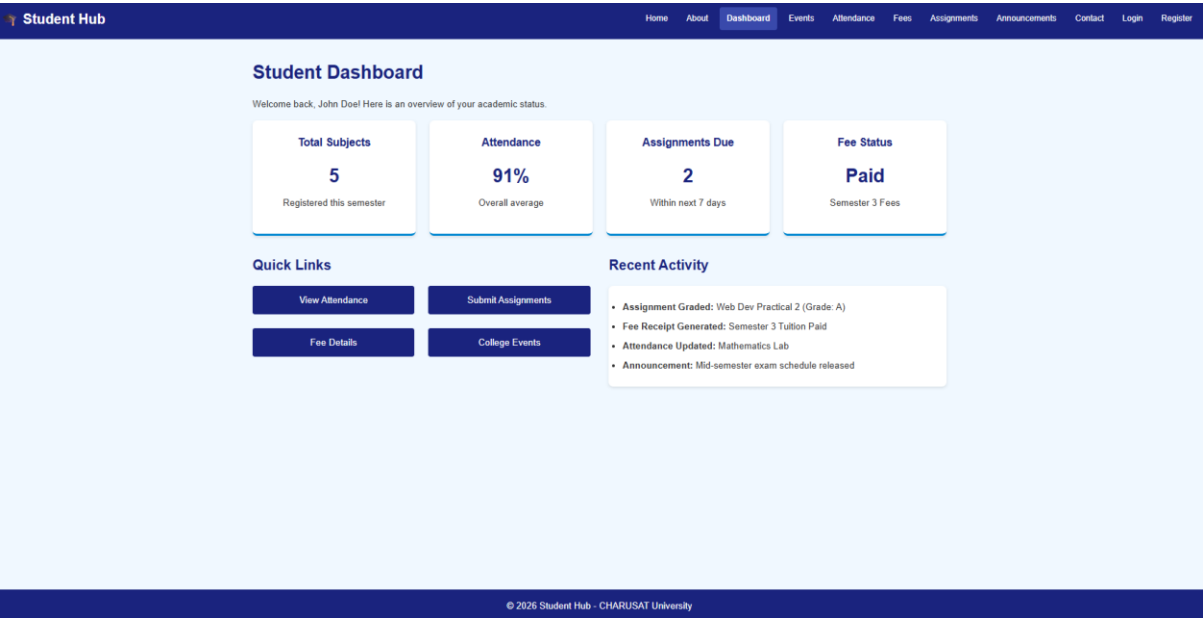
1. Home page:



2. About page:



3. Dashboard page:



4. Event page:

Student Hub

[Home](#)[About](#)[Dashboard](#)[Events](#)[Attendance](#)[Fees](#)[Assignments](#)[Announcements](#)[Contact](#)[Login](#)[Register](#)

Upcoming Events

Participate in technical, cultural, and sports events happening around the campus.

Technical

Tech Fest 2026
Date: 25-26 Sep 2026
Annual technical festival featuring hackathons, robotics, and project exhibitions.
[Register Now](#)

Competition

Coding Contest
Date: 15 Oct 2026
University-wide competitive programming contest. Open for all branches.
[Register Now](#)

Workshop

Workshop on AI
Date: 05 Nov 2026
A hands-on workshop on Artificial Intelligence and Machine Learning models.
[Register Now](#)

Sports

Annual Sports Day
Date: 12 Dec 2026
Inter-departmental sports tournament including cricket, volleyball, and athletics.
[Register Now](#)

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5. Attendance page:

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Attendance Record

Your attendance details for the current semester.

Subject Code	Subject Name	Total Lectures	Attended	Percentage	Status
CS301	Web Development	40	36	90%	Good
CS302	Data Structures & Algorithms (DSA)	50	44	88%	Good
MA303	Advanced Mathematics	40	38	95%	Excellent
CS304	Database Management	45	30	66%	Warning

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6. Fees page:

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HomeAboutDashboardEventsAttendanceFeesAssignmentsAnnouncementsContactLoginRegister

Fee Details

Review your academic fee payment history and pending dues.

Semester	Fee Type	Amount (INR)	Due Date	Payment Status
Semester 1	Tuition Fee	65,000	10 Jul 2025	Paid
Semester 2	Tuition Fee	65,000	15 Jan 2026	Paid
Semester 3	Tuition Fee	65,000	10 Jul 2026	Paid
Semester 4	Tuition Fee	65,000	15 Jan 2027	Pending

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7. Assignments page:

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HomeAboutDashboardEventsAttendanceFeesAssignmentsAnnouncementsContactLoginRegister

Assignments

Web Development - Practical 3

Due: 10 Aug 2026

Create a responsive UI using CSS Grid and Flexbox for the Student Hub portal.

Submit Work

DSA - Assignment 2

Due: 15 Aug 2026

Implement Binary Search Tree and its traversals in C++.

Submit Work

Mathematics - Problem Set 4

Due: 20 Aug 2026

Solve problems on probability distribution and linear algebra.

Submit Work

Database - E-R Diagram

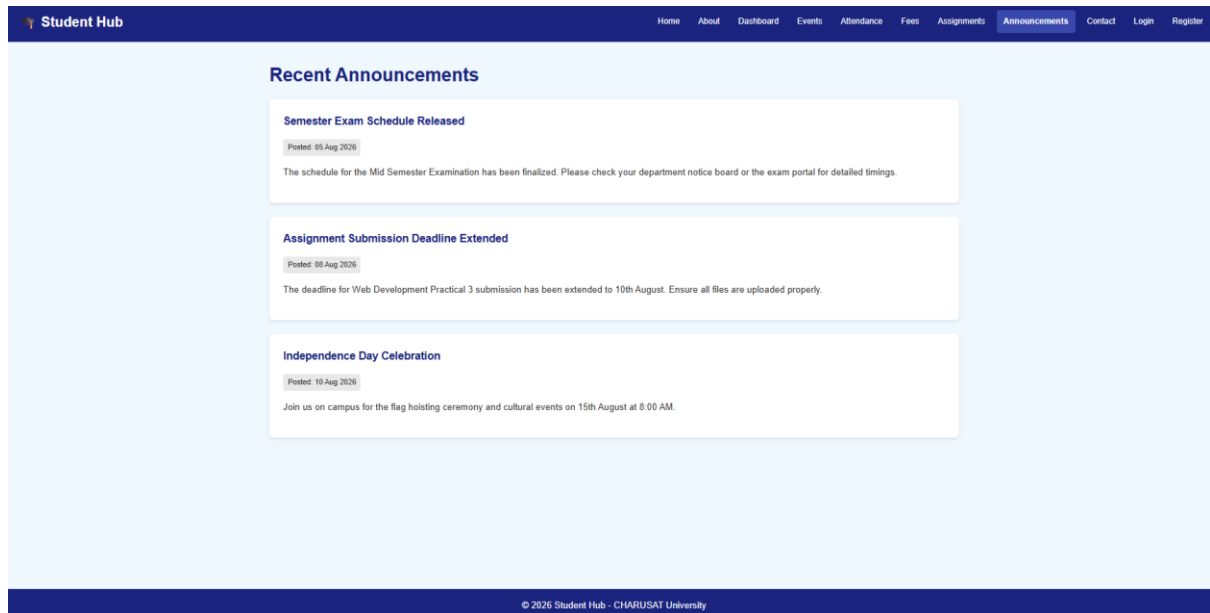
Due: 25 Aug 2026

Design an E-R diagram for a hospital management system.

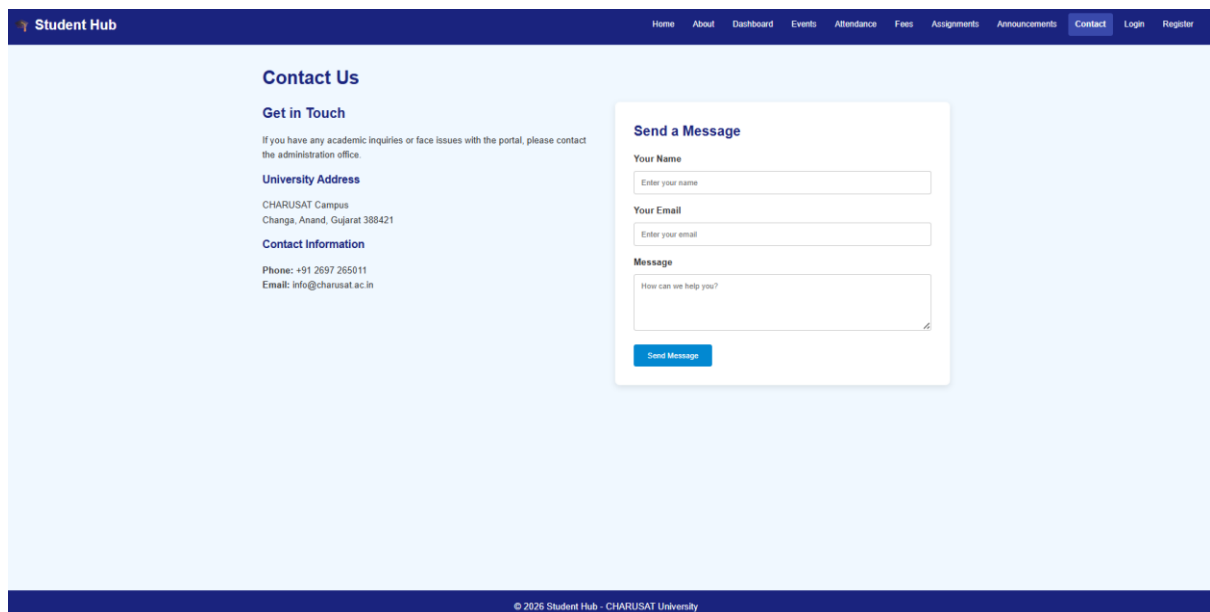
Submit Work

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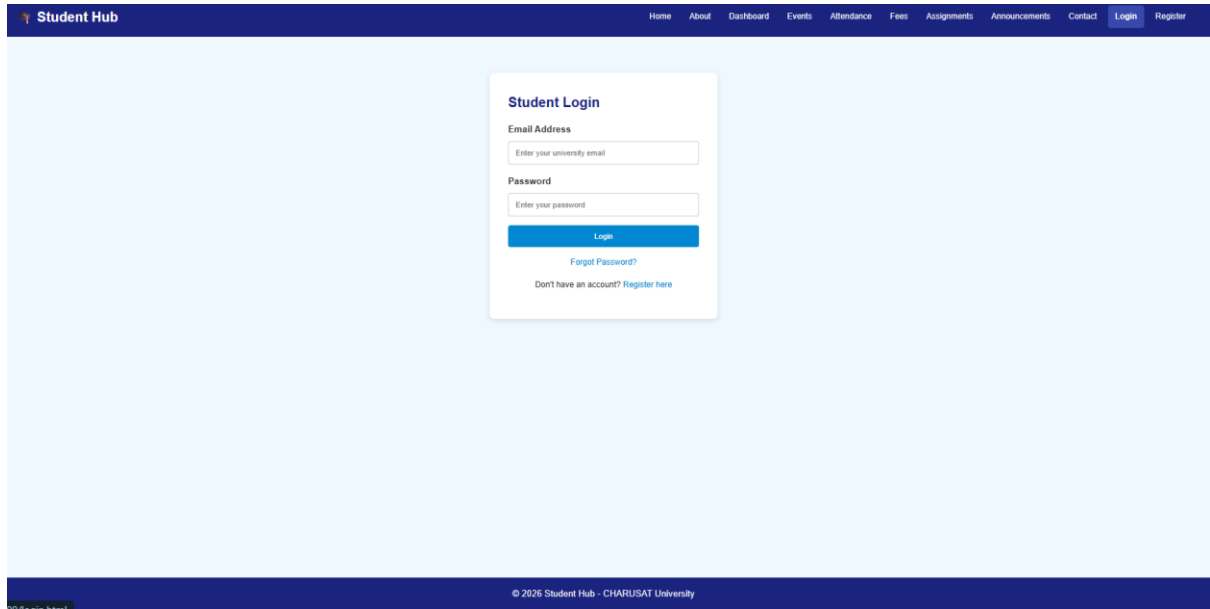
8. Announcements page:



9. Contact page:



10. Login page:



The screenshot shows the 'Student Login' page of the Student Hub. The page has a dark blue header with the 'Student Hub' logo and a navigation menu. The main content area is light blue and contains a white login form. The form has two input fields: 'Email Address' and 'Password'. Below the password field is a blue 'Login' button. There are also links for 'Forgot Password?' and 'Don't have an account? Register here'. The footer is dark blue and contains the copyright notice '© 2026 Student Hub - CHARUSAT University'.

Student Hub

Home About Dashboard Events Attendance Fees Assignments Announcements Contact **Login** Register

Student Login

Email Address

Password

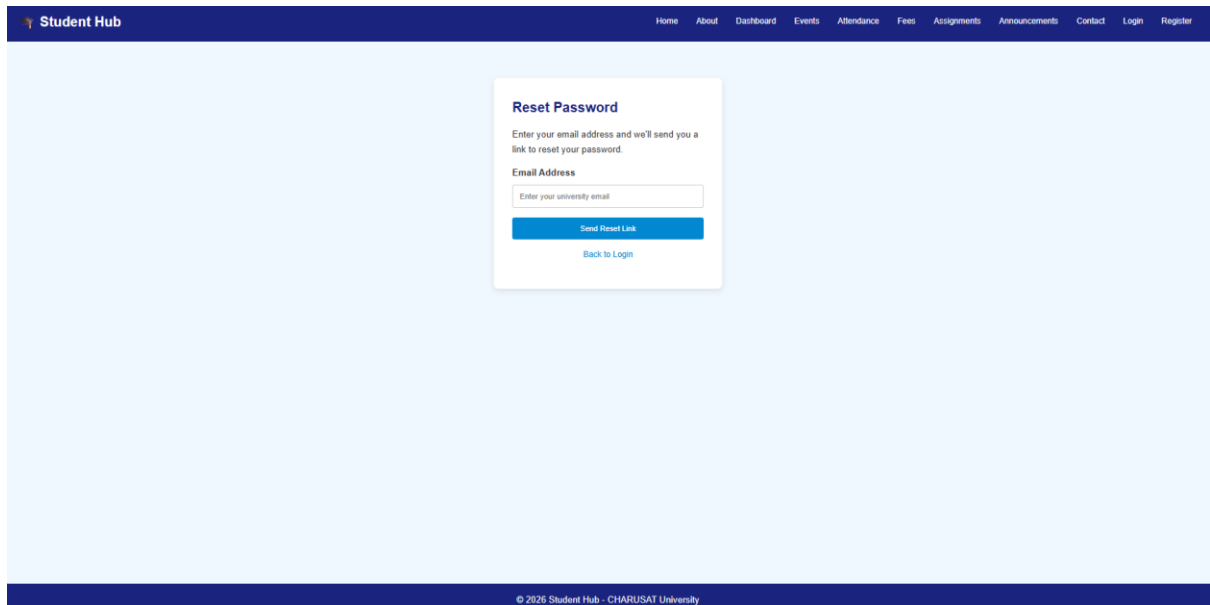
Login

[Forgot Password?](#)

Don't have an account? [Register here](#)

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11. Forget password page:



The screenshot shows the 'Reset Password' page of the Student Hub. The page has a dark blue header with the 'Student Hub' logo and a navigation menu. The main content area is light blue and contains a white form. The form has a heading 'Reset Password' and a paragraph 'Enter your email address and we'll send you a link to reset your password.' Below this is an 'Email Address' input field. There is a blue 'Send Reset Link' button and a link 'Back to Login'. The footer is dark blue and contains the copyright notice '© 2026 Student Hub - CHARUSAT University'.

Student Hub

Home About Dashboard Events Attendance Fees Assignments Announcements Contact Login Register

Reset Password

Enter your email address and we'll send you a link to reset your password.

Email Address

Send Reset Link

[Back to Login](#)

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12. Register page:

Student Hub

[Home](#) [About](#) [Dashboard](#) [Events](#) [Attendance](#) [Fees](#) [Assignments](#) [Announcements](#) [Contact](#) [Login](#) [Register](#)

Student Registration

Full Name

John Doe

University Email

example@charusat.edu.in

Mobile Number

1234567890

Password

Create a strong password

Register

Already have an account? [Login here](#)

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